Detecting Depression from Interview Speech

How the model improved over three versions, the results, and how the final model works

Final: 79.1% accuracy (cross-validation), 78.8% on 33 unseen people

The task: screen for depression from one interview

Input: audio of a 7-20 minute clinical interview

Output: is PHQ-8 >= 10 (clinically depressed)? yes / no

Data: DAIC-WOZ + E-DAIC (USC), the standard research benchmark

272

people interviewed

85

depressed (31%)

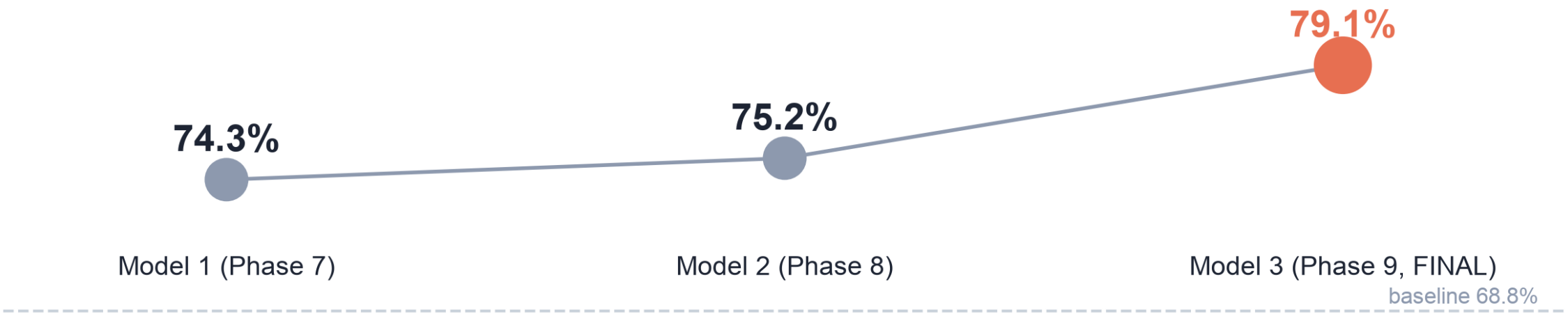
68.8%

accuracy of always guessing "not depressed" (the baseline)

Part 1: our progress in three models

Model 1 (Phase 7) -> Model 2 (Phase 8) -> Model 3 (Phase 9, final)

Progress at a glance



Model 1

Voice + text. Picks its best voice settings again in every training round.

Model 2

Same, but the 3 voice settings are fixed. 12 extra learning tricks tested: none helped.

Model 3 (final)

Model 2 + a local LLM that reads the transcript and rates the 8 PHQ-8 symptoms.

Model 1 (Phase 7): voice + text, 74.3%

MODEL 1



Result

- Accuracy 74.3% (baseline 68.8%)
- AUC 0.764, balanced accuracy 67.0%
- Catches 64% of depressed people (balanced threshold)

What held it back

- Voice alone carries a weak signal (AUC ~0.72)
- Text model only captures emotional tone, not symptoms
- Choosing among 60 settings each round adds noise

Model 2 (Phase 8): fixed voice settings, 75.2%

MODEL 2

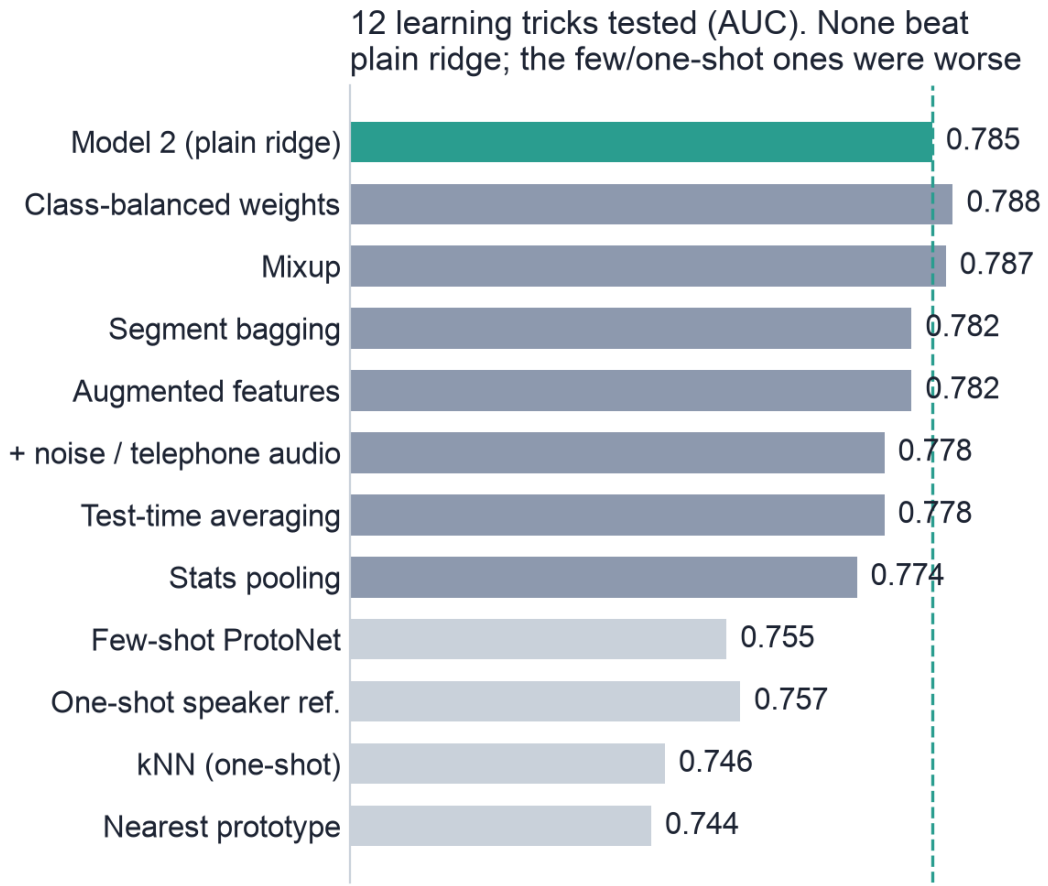
What changed

Always use the same 3 voice settings: wav2vec2 layer 6 and HuBERT layers 19 and 20, each with ridge regression.

Then we tested 12 learning tricks (teacher's ideas included)

Data augmentation, mixup, class weights, few-shot (prototypical network), one-shot (kNN, prototypes), speaker referencing, domain centring, statistics pooling.

Result: 75.2% accuracy, AUC 0.785. None of the tricks helped.



Model 3 (Phase 9, final): + LLM symptom scores, 79.1%

MODEL 3

Why we added it

- Model 2 showed new tricks do not help: we needed new information
- Our literature review ranked "LLM scores PHQ-8 items from the transcript" as the most promising next step
- People mention symptoms ("I can't sleep", "always tired"); the old text model ignored this

What changed

- Mistral-7B reads the transcript and rates each of the 8 PHQ-8 symptoms (0-3)
- Voice says HOW they speak; LLM says WHAT they report: the two add up
- **Accuracy 79.1%, AUC 0.843 (+0.057 over Model 2, statistically significant)**

What is inside each model

	Model 1 (Phase 7)	Model 2 (Phase 8)	Model 3 (FINAL)
Voice	best 3 of 60 settings, re-picked each round	fixed: wav2vec2 L6, HuBERT L19 + L20	same as Model 2
Text	emotion-RoBERTa	emotion-RoBERTa	emotion-RoBERTa
Symptom scores (LLM)	no	no	**YES: Mistral-7B, 8 PHQ-8 items**
Combiner	logistic regression	logistic regression	logistic regression
Accuracy, cross-validation	74.3%	75.2%	**79.1%**
Accuracy, 33 unseen	72.7%	72.7%	**78.8%**

Only one ingredient is new in Model 3, so the gain comes from the LLM symptom scores.

Part 2: results

Test A: cross-validation on all 272 people

Test B: train on 239, test once on 33 unseen people

Two honest tests, no data leakage

A Test A: cross-validation

All 272 people. Train on 4/5, test on the other 1/5, rotate until everyone is tested; repeat with 5 different splits.

Each person is predicted by a model that never saw them.

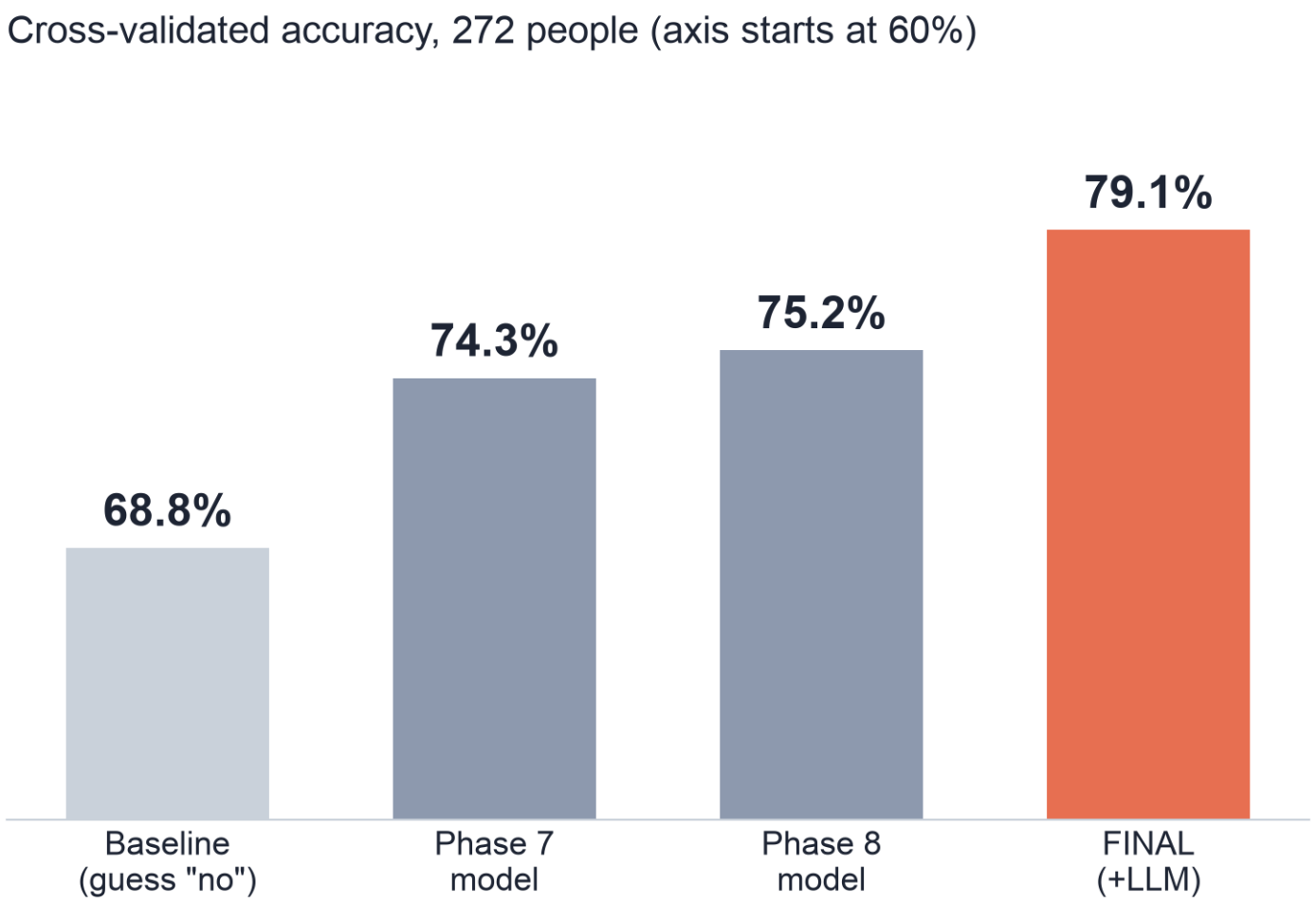
B Test B: unseen holdout

Train only on 239 people. Test once on 33 people kept aside in the "unseen" folder.

All settings (strengths, combiner, threshold) chosen on the 239 only.

Rules we kept: nobody appears in both training and test; the LLM never sees a label; interviewer questions removed from transcripts; plan and adoption rule written and pushed to GitHub before running.

Result A: accuracy rises from 74.3% to 79.1%



0.843

AUC (Model 1: 0.764)

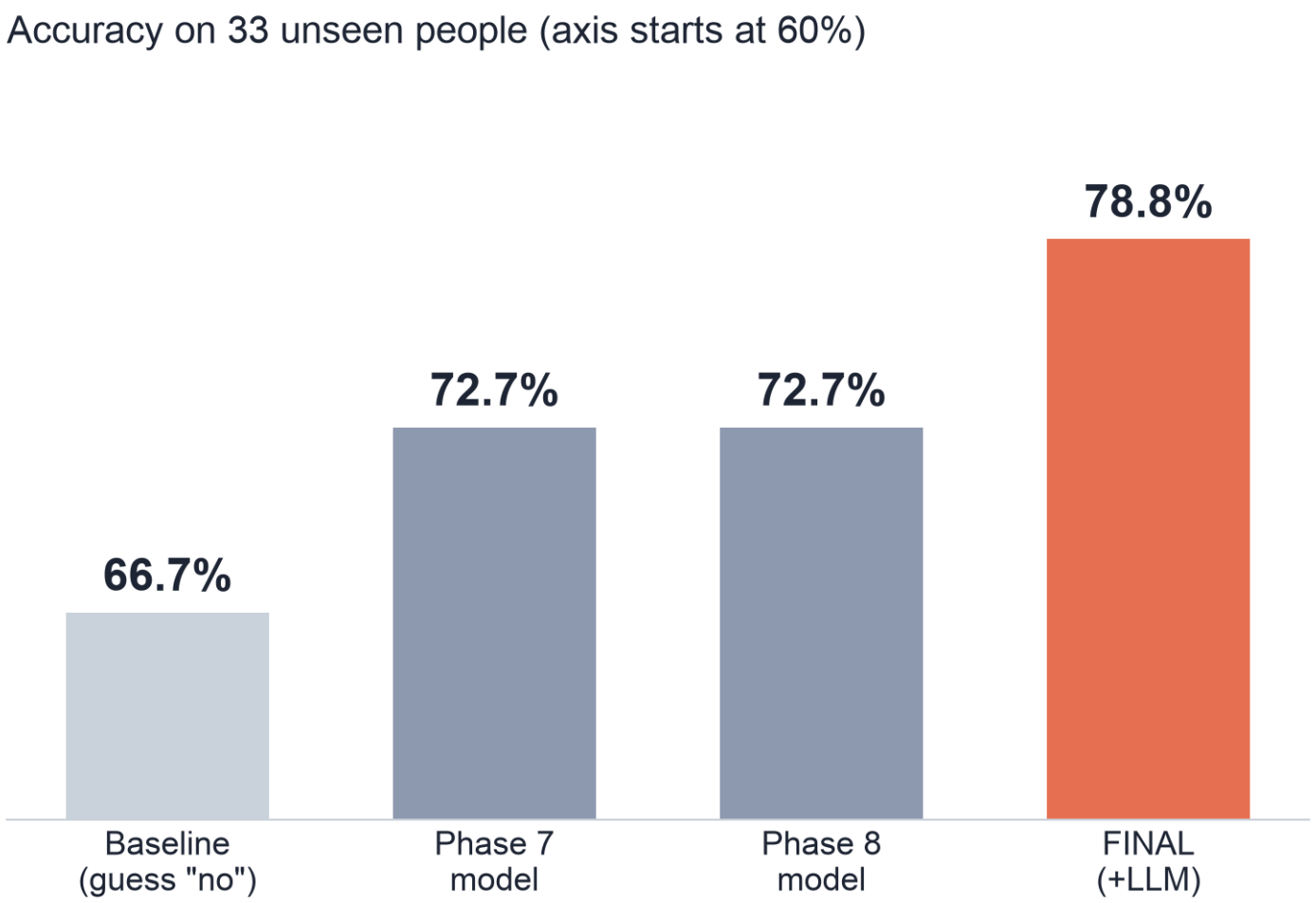
72.4%

balanced accuracy (Model 1: 67.0%)

77%

of depressed people caught at the balanced threshold (Model 1: 64%)

Result B: 26 of 33 unseen people correct



0.868

AUC (Model 1: 0.777)

21 / 22

healthy people correctly cleared

9 / 11

depressed caught at the balanced threshold (Model 1: 7)

All results side by side

Metric	Model 1	Model 2	Model 3 (FINAL)
Test A: cross-validation, 272 people (baseline 68.8%)			
Accuracy	74.3%	75.2%	**79.1%**
Balanced accuracy	67.0%	68.3%	**72.4%**
AUC	0.764	0.785	**0.843**
MCC / macro-F1	0.37 / 0.68	0.39 / 0.69	**0.49 / 0.74**
Test B: train 239, test 33 unseen (baseline 66.7%)			
Accuracy	72.7% (24/33)	72.7% (24/33)	**78.8% (26/33)**
AUC	0.777	0.785	**0.868**
Depressed caught (balanced thr.)	7 / 11	7 / 11	**9 / 11**

Model 3 vs Model 1: AUC +0.079 [95% CI +0.035, +0.124]. Model 3 vs Model 2: +0.057 [+0.015, +0.102]. Both significant.

Screenshot: the results as printed

Terminal: results

```
$ python run_project.py --report

Phase 9: nested cross-validation, N = 272 (baseline 68.8%)
   model                              AUC      acc  bal-acc   sens   spec
   P4  (74.3% model)                  0.764    74.3%    67.0%    48%    87%
   R0_voice_text  (75% model)         0.785    75.2%    68.3%    50%    87%
   F1_voice_text_llm  (FINAL)         0.843    79.1%    72.4%    55%    90%

Holdout: train 239 -> test the 33 unseen (baseline 66.7%)
   model                              AUC      acc  bal-acc   sens   spec
   P4  (74.3% model)                  0.777    72.7%    68.2%    55%    82%
   R0_voice_text  (75% model)         0.785    72.7%    70.5%    64%    77%
   F1_voice_text_llm  (FINAL)         0.868    78.8%    70.5%    45%    95%

Final saved model: models/final_phase9_f1.joblib  (trained on all 272)
```

Top block: Test A, cross-validation on 272.

Bottom block: Test B, trained on 239, tested on 33 unseen.

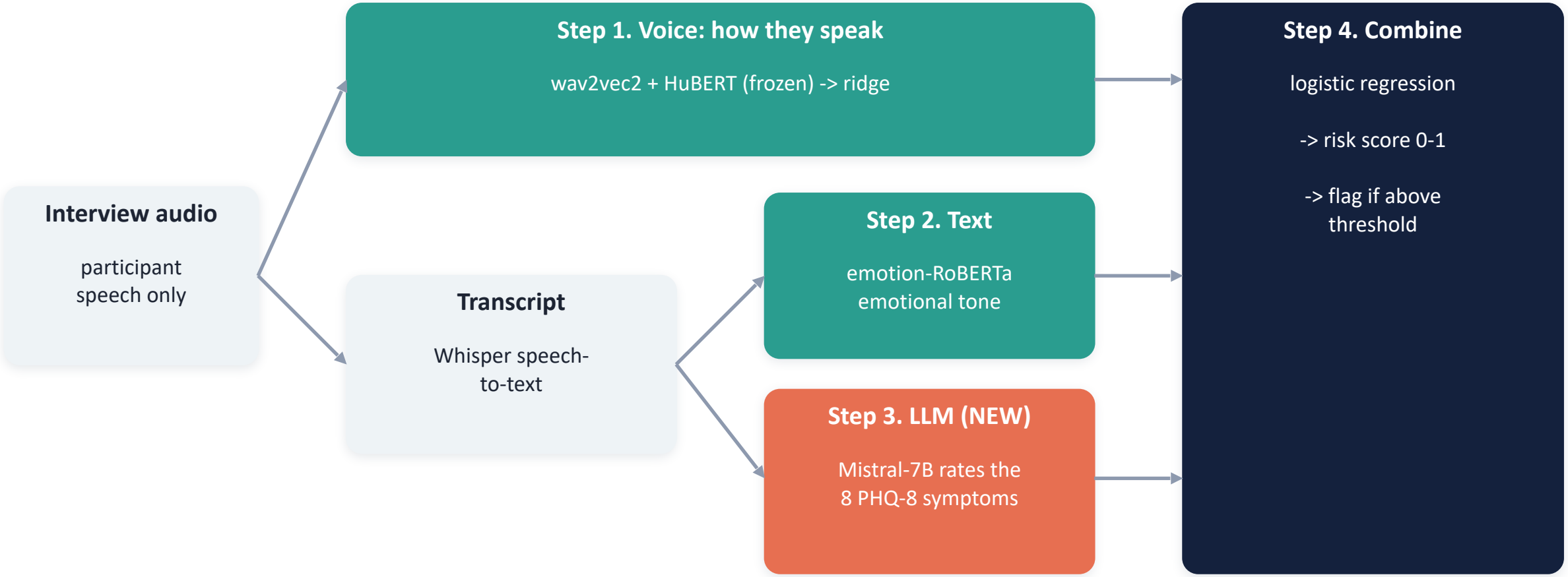
P4 = Model 1, R0 = Model 2, F1 = Model 3 (orange).

acc = accuracy, bal-acc = balanced accuracy, sens = depressed caught, spec = healthy cleared.

Part 3: the final model, step by step

Step 1 voice | Step 2 text | Step 3 LLM symptom scores | Step 4 combine

The final model: four steps



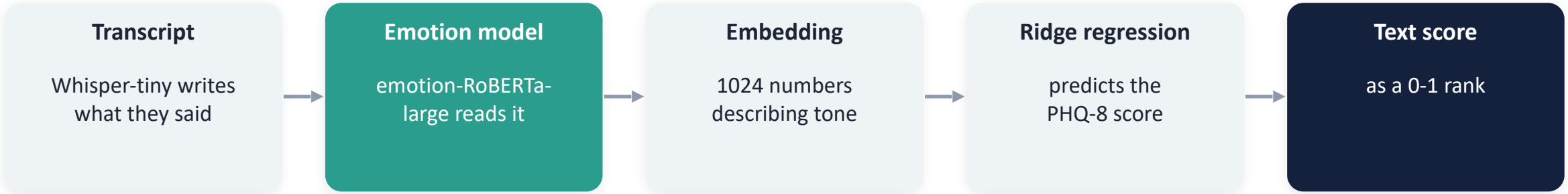
Each step gives its own score from 0 to 1; step 4 merges the three into one risk score.

Steps 1 and 2: voice and text scores

Step 1: Voice (how they speak)



Step 2: Text (emotional tone of the words)



The encoders are pre-trained and frozen; we only train the small ridge regressions (strength chosen by inner cross-validation).

Step 3: how the LLM scores symptoms

Transcript
participant's words only

→

Prompt
"Rate each PHQ-8 item 0-3"

→

Mistral-7B
4-bit, runs offline on the laptop

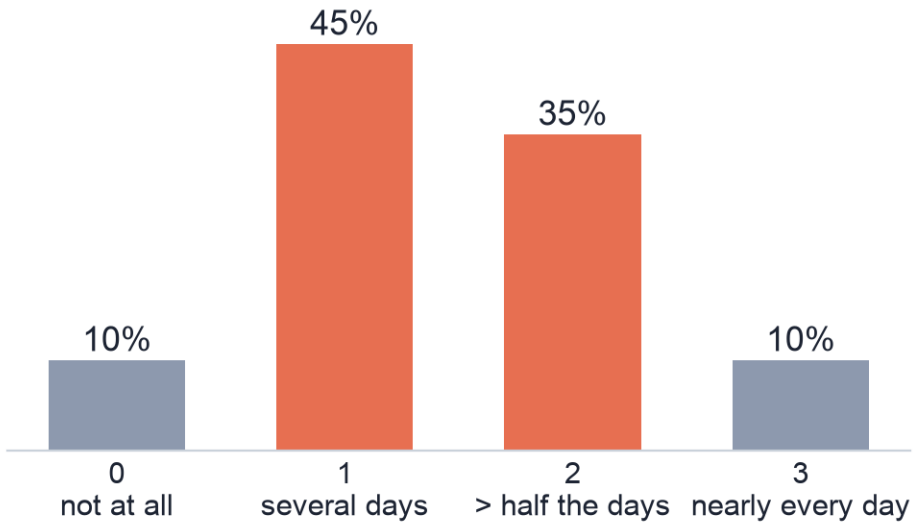
→

8 symptom scores
one number 0-3 per symptom

→

Ridge -> LLM score
learns which symptoms matter, 0-1 rank

Example: LLM's probabilities for "sleep"



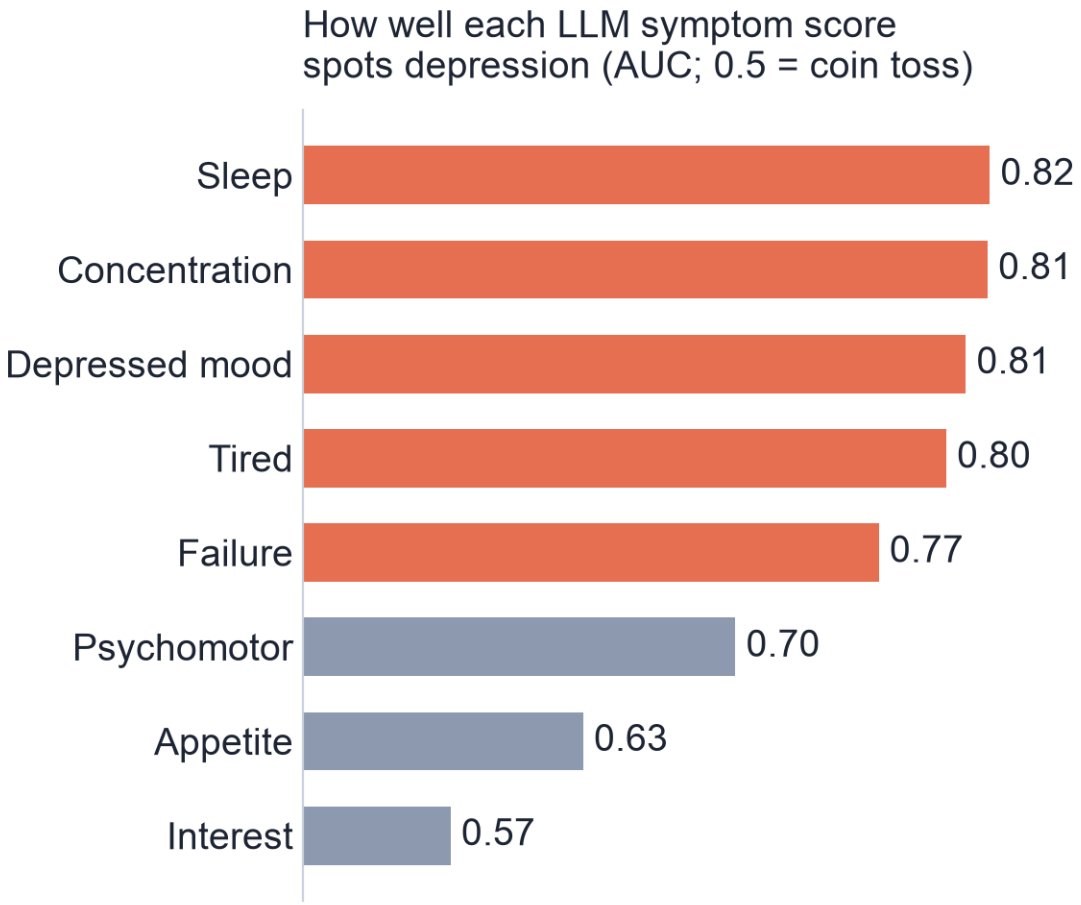
Not just the answer: the probabilities

For each symptom we read how likely the model thinks 0, 1, 2 and 3 are, and take the average:

`0 x 0.10 + 1 x 0.45 + 2 x 0.35 + 3 x 0.10 = 1.45`

A score of 1.45 keeps the model's uncertainty, which works better than a hard "1".

Step 3: which symptoms the LLM detects best



What this shows

- Sleep, concentration, mood and tiredness each spot depression well on their own (AUC ~0.8)
- Interest is weak: the LLM gives almost everyone the same score
- All 8 together, with no training: AUC 0.82
- Ridge then learns the best weighting: AUC 0.83

Step 4: combining the three scores

How much each part counts (logistic-regression weights)



The LLM counts about twice as much as voice or text.

Two decision thresholds

Risk score 0 to 1 from the three weighted parts.

Max-accuracy: flag if > 0.542
fewest mistakes overall (79% accuracy), but catches only about half of depressed people

Balanced: flag if > 0.196
catches 77% of depressed people; better for screening

Training the final model and using it

Terminal: predict_audio.py on one recording

```
$ python predict_audio.py NEW_AUDIO_DAIC/606_AUDIO_interviewee.wav --domain avatar

Scoring PHQ-8 items with the local LLM ...
LLM PHQ-8 item estimates (0-3): interest 0.1, depressed 0.0, sleep 0.0,
   tired 0.4, appetite 0.0, failure 0.0, concentration 1.3, psychomotor 0.0 | sum 1.9

Depression risk score: 0.036  (0 = low, 1 = high)
  Max-accuracy threshold 0.542: not flagged
  Balanced threshold     0.196: not flagged

Model: Phase 9 F1, nested-CV AUC 0.84 / accuracy 79%; unseen-33 holdout 78.8%.
(True PHQ-8 of participant 606 = 5: not depressed -> correct)
```

Trained on all 272 people with exactly the steps that were tested.

One command: audio in, risk score out, with the 8 symptom ratings shown.

Matches the training pipeline exactly (score 0.036 for this person).

Saved as models/final_phase9_f1.joblib. Runs offline on an 8 GB MacBook in about 1-3 minutes. Screening aid only, not a diagnosis.

Why we trust it, and its limits

Why it is real

- Both tests agree: 79.1% and 78.8%
- Without the 17 people who say "I am depressed": AUC still 0.835
- Works for both interview setups (AUC 0.84 / 0.86) and both genders
- LLM never sees labels; nothing tuned on test people

Limits

- 33 unseen people is small: 1 person = 3 points
- The 33 were opened once before (Oct 19), so a repeat check
- At the max-accuracy threshold only about half of depressed people are caught
- Research prototype, not a diagnosis

74.3% -> 79.1% accuracy

Model 1 to Model 2: fixed voice settings, +0.9 points. Model 2 to Model 3: added local LLM symptom scoring, +3.9 points.

Final model: 79.1% (cross-validation, AUC 0.84) and 78.8% on 33 unseen people (AUC 0.87), against a 68.8% / 66.7% baseline.